

NAME

lcalib -- List information about the spectral calibration data

USAGE

lcalib option star_name

PARAMETERS**option**

Chooses calibration data to be listed. Option may be: "bands" to list the bandpasses at each wavelength, "ext" to list the extinction at each wavelength, "mags", "fnu", or "flam" to list the magnitude, or flux of the star (selected by the star_name parameter) at each wavelength, or "stars" to list the star names available in the calibration directory.

star_name

Selects which star's magnitude list is chosen if the option parameter is "mags", "fnu", "flam", or "bands". Also if '?' a list of available stars in the specified calibration directory is given.

The following three queried parameters apply if the selected calibration file is for a blackbody. See **standard** for further details.

mag The magnitude of the observed star in the band given by the *magband* parameter. If the magnitude is not in the same band as the blackbody calibration file then the magnitude may be converted to the calibration band provided the "params.dat" file containing relative magnitudes between the two bands is in the calibration directory

magband

The standard band name for the input magnitude. This should generally be the same band as the blackbody calibration file. If it is not the magnitude will be converted to the calibration band.

teff The effective temperature (deg K) or the spectral type of the star being calibrated. If a spectral type is specified a "params.dat" file must exist in the calibration directory. The spectral types are specified in the same form as in the "params.dat" file. For the standard blackbody calibration directory the spectral types are specified as A0I, A0III, or A0V, where A can be any letter OBAFGKM, the single digit subclass is between 0 and 9, and the luminosity class is one of I, III, or V. If no luminosity class is given it defaults to dwarf.

extinction

Extinction file. The current standard extinction files:

```
onedstds$kpnoextinct.dat - KPNO standard extinction
onedstds$ctioextinct.dat - CTIO standard extinction
```

caldir

Calibration directory containing standard star data. The directory name must end with /. The current calibration directories available in the onedstds\$ may be listed with the command:

```
cl> page onedstds$README
```

fnuzero= 3.68e-20

The absolute flux per unit frequency at a magnitude of zero. This is to convert the calibration

magnitudes to absolute flux by the formula

$$\text{Flux} = \text{fnuzero} * 10. ** (-0.4 * \text{magnitude})$$

The flux units are also determined by this parameter. However, the frequency to wavelength interval conversion assumes frequency in hertz. The default value is based on a calibration of Vega at 5556 Angstroms of $3.52\text{e-}20$ ergs/cm²/s/hz for a magnitude of 0.048. This default value is that used in earlier versions of this task which did not allow the user to change this calibration.

DESCRIPTION

LCALIB provides a means of checking the flux calibration data. The calibration data consists of extinction, bandpasses, and stellar magnitudes.

The extinction is given in an extinction file consisting of lines with wavelength and extinction. The wavelengths must be order in increasing wavelength and the wavelengths must be in Angstroms. There are two standard extinction files currently available, "onedstds\$kpnoextinct.dat", and "onedstds\$ctioextinct.dat".

The standard star data are in files in a calibration directory specified with the parameter *caldir*. A standard star file is selected by taking the star name given, by the parameter *star_name*, removing blanks, +’s and -’s, appending ".dat", and converting to lower case. This file name is appended to the specified calibration directory. A calibration file consists of lines containing a stellar magnitude, and a bandpass full width. The wavelengths are in Angstroms. Comment lines beginning with # may be included in the file. The star names printed by this task are just the first line of each file in the calibration directory with the first character (#) removed. The calibration files may be typed, copied, and printed. **Lcalib** may also be used to list data from the calibration files.

EXAMPLES

```
# List the extinction table
cl> lcalib ext
# Plot the extinction table
cl> lcalib ext | graph
# Plot the energy distribution
cl> lcalib mags "bd+28 4211" | graph
# List the names of all the stars
cl> lcalib stars caldir=onedstds$irscal/
# As above but for IIDS file
cl> lcalib stars calib_file=onedstds$iidscal/
```

REVISIONS

LCALIBV2.10

This task has a more compact listing for the "stars" option and allows paging a list of stars when the star name query is not recognized.

SEE ALSO

standard, sensfunc, onedstds\$README